

*Las def estan abaso de todo

Ejercicio 9

De acuerdo a las definiciones de las funciones para árboles ternarios de más arriba, se pide demostrar lo siguiente:

$$\forall t :: AT\ a. \forall x :: a. (elem\ x\ (preorder\ t) = elem\ x\ (postorder\ t))$$

Por principio de inducción sobre Árboles Ternarios. Queremos probar que:

$$\forall t :: AT\ a. \forall x :: a. (elem\ x\ (preorder\ t) = (elem\ x\ (postorder\ t)))$$

$$P(t) = \forall x :: a. (elem\ x\ (preorder\ t) = (elem\ x\ (postorder\ t)))$$

Caso Base:

$$t = Nil$$

$$P(Nil) = \forall x :: a. (elem\ x\ (preorder\ Nil) = (elem\ x\ (postorder\ Nil)))$$

$$(elem\ x\ (preorder\ Nil)) \overset{\{preorder\}}{=} (elem\ x\ (foldAT\ []\ (\backslash\ r\ i\ m\ d \rightarrow [r] ++ i ++ m ++ d)\ Nil)) =$$

$$\overset{\{foldAT\}}{=} (elem\ x\ []) \overset{\{elem\}}{=} False$$

$$(elem\ x\ (postorder\ Nil)) \overset{\{postorder\}}{=} (elem\ x\ (foldAT\ []\ (\backslash\ r\ i\ m\ d \rightarrow i ++ m ++ d ++ [r])\ Nil)) =$$

$$\overset{\{foldAT\}}{=} (elem\ x\ []) \overset{\{elem\}}{=} False$$

$$False = False \checkmark$$

Caso Inductivo:

$$\forall i :: AT\ a. \forall m :: AT\ a. \forall d :: AT\ a. \forall r :: a$$

$$\underbrace{(P(i) \wedge P(m) \wedge P(d))}_{HI} \Rightarrow \underbrace{P(Tern\ r\ i\ m\ d)}_{TI}$$

$$(P(i) = \forall x :: a. (elem\ x\ (preorder\ i) = (elem\ x\ (postorder\ i)))) \wedge$$

$$(P(m) = \forall x :: a. (elem\ x\ (preorder\ m) = (elem\ x\ (postorder\ m)))) \wedge$$

$$(P(d) = \forall x :: a. (elem\ x\ (preorder\ d) = (elem\ x\ (postorder\ d))))$$

Queremos probar que

$$\rightarrow \forall x :: a.$$

$$(elem\ x\ (preorder\ (Term\ r\ i\ m\ d))) = (elem\ x\ (postorder\ (Term\ r\ i\ m\ d)))$$

$$(\text{elem } x (\text{preorder } (\text{Term } r \ i \ m \ d))) \stackrel{\{\text{Preorder}\}}{=} \dots$$

$$(\text{elem } x (\text{foldAT } [] (\lambda r \ i \ m \ d \rightarrow [r] ++ i ++ m ++ d) (\text{Term } r \ i \ m \ d))) \stackrel{\{\text{foldAT}\}}{=} \text{rec} = \text{foldAT } [] (\lambda r \ i \ m \ d \rightarrow [r] ++ i ++ m ++ d)$$

$$(\text{elem } x ((\lambda r \ i \ m \ d \rightarrow [r] ++ i ++ m ++ d) \text{r} (\text{rec } i) (\text{rec } m) (\text{rec } d))) \stackrel{B}{=} \dots$$

$$(\text{elem } x ((\lambda i \ m \ d \rightarrow [r] ++ i ++ m ++ d) (\text{rec } i) (\text{rec } m) (\text{rec } d))) \stackrel{B}{=} \dots$$

$$(\text{elem } x ((\lambda m \ d \rightarrow [r] ++ (\text{rec } i) ++ m ++ d) (\text{rec } m) (\text{rec } d))) \stackrel{B}{=} \dots$$

$$(\text{elem } x ((\lambda d \rightarrow [r] ++ (\text{rec } i) ++ (\text{rec } m) ++ d) (\text{rec } d))) \stackrel{B}{=} \dots$$

$$(\text{elem } x ([r] ++ (\text{rec } i) ++ (\text{rec } m) ++ (\text{rec } d))) \rightarrow \text{Por Ejercicio de la guía 2 (3.3)}$$

$$(\text{elem } x (\text{r} : (\text{rec } i) ++ (\text{rec } m) ++ (\text{rec } d))) \stackrel{\{\text{elem 1}\}}{=} \dots$$

$$x == r \parallel \text{elem } x ((\text{rec } i) ++ (\text{rec } m) ++ (\text{rec } d))$$

$$(\text{elem } x (\text{postorder } (\text{Term } r \ i \ m \ d))) \stackrel{\{\text{postorder}\}}{=} \dots$$

$$(\text{elem } x (\text{foldAT } [] (\lambda r \ i \ m \ d \rightarrow i ++ m ++ d ++ [r]) (\text{Term } r \ i \ m \ d))) \stackrel{\{\text{foldAT}\}}{=} \text{rec}_2 = \text{foldAT } [] (\lambda r \ i \ m \ d \rightarrow i ++ m ++ d ++ [r])$$

$$(\text{elem } x ((\lambda r \ i \ m \ d \rightarrow i ++ m ++ d ++ [r]) \text{r} (\text{rec}_2 \ i) (\text{rec}_2 \ m) (\text{rec}_2 \ d))) \stackrel{B}{=} \dots$$

$$(\text{elem } x ((\lambda i \ m \ d \rightarrow i ++ m ++ d ++ [r]) (\text{rec}_2 \ i) (\text{rec}_2 \ m) (\text{rec}_2 \ d))) \stackrel{B}{=} \dots$$

$$(\text{elem } x ((\lambda m \ d \rightarrow (\text{rec}_2 \ i) ++ m ++ d ++ [r]) (\text{rec}_2 \ m) (\text{rec}_2 \ d))) \stackrel{B}{=} \dots$$

$$(\text{elem } x ((\lambda d \rightarrow (\text{rec}_2 \ i) ++ (\text{rec}_2 \ m) ++ d ++ [r]) (\text{rec}_2 \ d))) \stackrel{B}{=} \dots$$

$$(\text{elem } x ((\text{rec}_2 \ i) ++ (\text{rec}_2 \ m) ++ (\text{rec}_2 \ d) ++ [r])) \stackrel{\text{Lemma 1}}{=} \dots$$

$$((\text{elem } x \text{rec}_2 \ i) \parallel (\text{elem } x ((\text{rec}_2 \ m) ++ (\text{rec}_2 \ d) ++ [r]))) \stackrel{\text{Lemma 1}}{=} \dots$$

$$(\text{elem } x \text{rec}_2 \ i) \parallel ((\text{elem } x (\text{rec}_2 \ m) \parallel (\text{elem } x ((\text{rec}_2 \ d) ++ [r]))) \stackrel{\text{Lemma 1}}{=} \dots$$

$$(\text{elem } x (\text{rec}_2 \ i) \parallel (\text{elem } x (\text{rec}_2 \ m) \parallel (\text{elem } x (\text{rec}_2 \ d) \parallel (\text{elem } x [r]))) \stackrel{\{\text{elem 1}\}}{=} \dots$$

$$(\text{elem } x (\text{rec}_2 \ i) \parallel (\text{elem } x (\text{rec}_2 \ m) \parallel (\text{elem } x (\text{rec}_2 \ d) \parallel (x == r \parallel \text{elem } x []))) \stackrel{\{\text{elem 0}\}}{=} \dots$$

$$(\text{elem } x (\text{rec}_2 \ i) \parallel (\text{elem } x (\text{rec}_2 \ m) \parallel (\text{elem } x (\text{rec}_2 \ d) \parallel (x == r \parallel \text{False}))) = \dots$$

$$(\text{elem } x (\text{rec}_2 \ i) \parallel (\text{elem } x (\text{rec}_2 \ m) \parallel (\text{elem } x (\text{rec}_2 \ d) \parallel x == r))$$

Por extensionalidad booleanos tenemos 2 casos

- $x == r = \text{True}$

- $x == r = \text{False}$

Caso $x == r = \text{True}$

Lado izquierdo del igual:

$$\text{True} \parallel \text{elem } x \left((\text{rec } i) ++ (\text{rec } m) ++ (\text{rec } d) \right) = \text{True}$$

Lado derecho del igual:

$$(\text{True} = \text{True}) = \text{True} \quad \checkmark$$

$$(\text{elem } x (\text{rec } i)) \parallel (\text{elem } x (\text{rec } m)) \parallel (\text{elem } x (\text{rec } d)) \parallel \text{True} = \text{True}$$

Caso $x == r = \text{False}$

Lado izquierdo del igual:

$$\text{False} \parallel \text{elem } x \left((\text{rec } i) ++ (\text{rec } m) ++ (\text{rec } d) \right) =$$

$$\text{elem } x \left((\text{rec } i) ++ (\text{rec } m) ++ (\text{rec } d) \right) \stackrel{\text{Lema 1}}{=}$$

$$(\text{elem } x (\text{rec } i)) \parallel (\text{elem } x ((\text{rec } m) ++ (\text{rec } d))) \stackrel{\text{Lema 1}}{=}$$

$$(\text{elem } x (\text{rec } i)) \parallel (\text{elem } x (\text{rec } m)) \parallel (\text{elem } x (\text{rec } d)) =$$

$$(\text{elem } x (\text{foldAT } [] (\lambda r i m d \rightarrow [r] ++ i ++ m ++ d) i)) \parallel$$

$$(\text{elem } x (\text{foldAT } [] (\lambda r i m d \rightarrow [r] ++ i ++ m ++ d) m)) \parallel$$

$$(\text{elem } x (\text{foldAT } [] (\lambda r i m d \rightarrow [r] ++ i ++ m ++ d) d)) \stackrel{\text{Preorder}}{=}$$

$$(\text{elem } x (\text{preorder } i)) \parallel (\text{elem } x (\text{preorder } m)) \parallel (\text{elem } x (\text{preorder } d))$$

Lado derecho del igual:

$$(\text{elem } x (\text{rec } i)) \parallel (\text{elem } x (\text{rec } m)) \parallel (\text{elem } x (\text{rec } d)) \parallel \text{False} =$$

$$(\text{elem } x (\text{rec } i)) \parallel (\text{elem } x (\text{rec } m)) \parallel (\text{elem } x (\text{rec } d)) =$$

$$(\text{elem } x (\text{foldAT } [] (\lambda r i m d \rightarrow i ++ m ++ d ++ [r]) i)) \parallel$$

$$(\text{elem } x (\text{foldAT } [] (\lambda r i m d \rightarrow i ++ m ++ d ++ [r]) m)) \parallel$$

$$(\text{elem } x (\text{foldAT } [] (\lambda r i m d \rightarrow i ++ m ++ d ++ [r]) d)) \stackrel{\text{Postorder}}{=}$$

$$(\text{elem } x \text{ (postorder } i)) \parallel (\text{elem } x \text{ (postorder } m)) \parallel (\text{elem } x \text{ (postorder } d))$$

Por la HI podemos igualar tanto el lado izquierdo como el lado derecho.

$$\begin{array}{ccc}
 (\text{elem } x \text{ (preorder } i)) \parallel (\text{elem } x \text{ (preorder } m)) \parallel (\text{elem } x \text{ (preorder } d)) & \stackrel{\text{HI}}{=} & \\
 \downarrow & & \downarrow \\
 (\text{elem } x \text{ (postorder } i)) \parallel (\text{elem } x \text{ (postorder } m)) \parallel (\text{elem } x \text{ (postorder } d)) & &
 \end{array}$$

Por así decirlo en este paso aplicamos la HI 3 veces (una $\forall$ cada elemento).

Como en ambos lados llegamos a lo mismo queda probado para $P(\text{Term } r \ i \ m \ d)$

$$(\underbrace{P(\text{Nil})}_{\checkmark}) \wedge \forall i, m, d :: \text{ATA} . \forall r :: a . \underbrace{(P(i) \wedge P(m) \wedge P(d))}_{\checkmark} \Rightarrow P(\text{Term } r \ i \ m \ d)) \quad \checkmark$$

Lema 1:

$$\forall x_s :: [a] . \forall y_s :: [a] . \forall e :: a .$$

$$(\text{elem } e \text{ (} x_s ++ y_s)) = \text{elem } e \ x_s \parallel \text{elem } e \ y_s$$

Por inducción sobre listas en x_s , queremos probar que:

$$P(x_s) = \forall e :: a . \forall y_s :: [a] . (\text{elem } e \text{ (} x_s ++ y_s)) = \text{elem } e \ x_s \parallel \text{elem } e \ y_s$$

Caso Base: $x_s = []$

$$P([]) = \forall e :: a . \forall y_s :: [a] . (\text{elem } e \text{ (} [] ++ y_s)) = \text{elem } e \ [] \parallel \text{elem } e \ y_s$$

Lado izquierdo:

$$\text{elem } e \text{ (} [] ++ y_s) \stackrel{\text{def } ++}{=}$$

$$\text{elem } e \text{ (foldr (:) y_s [])} \stackrel{\text{def foldr D}}{=}$$

$$\text{elem } e \ y_s$$

Lado derecho:

$$\text{elem } e \ [] \parallel \text{elem } e \ y_s \stackrel{\text{def elem D}}{=} \text{False} \parallel \text{elem } e \ y_s = \text{elem } e \ y_s$$

Como el lado izquierdo y derecho a lo mismo, el caso base se cumple.

Caso Inductivo:

$$\forall x :: a. \forall xs :: [a]. \underbrace{P(xs)}_{HI} \Rightarrow \underbrace{P(x:xs)}_{T.I.}$$

$$P(xs) = (\text{elem } e \ (xs ++ ys)) = \text{elem } e \ xs \ \parallel \ \text{elem } e \ ys$$

$$P(x:xs) = (\text{elem } e \ ((x:xs) ++ ys)) = \text{elem } e \ (x:xs) \ \parallel \ \text{elem } e \ ys$$

Lado izquierdo:

$$(\text{elem } e \ ((x:xs) ++ ys)) \stackrel{\text{def } ++}{=} \text{elem } e \ (\text{foldr } (:) \ ys \ (x:xs))$$

$$\text{elem } e \ (\text{foldr } (:) \ ys \ (x:xs)) \stackrel{\text{def foldr 1}}{=} \text{elem } e \ ((:) \ x \ (\text{foldr } (:) \ ys \ xs))$$

$$\text{elem } e \ ((:) \ x \ (\text{foldr } (:) \ ys \ xs)) =$$

$$\text{elem } e \ (x : (\text{foldr } (:) \ ys \ xs)) \stackrel{\text{def } ++}{=} \text{elem } e \ (x : (xs ++ ys))$$

$$\text{elem } e \ (x : (xs ++ ys)) \stackrel{\text{def elem 1}}{=} \text{elem } e \ x \ \parallel \ \text{elem } e \ (xs ++ ys)$$

$$\text{elem } e \ x \ \parallel \ \text{elem } e \ (xs ++ ys) \stackrel{HI}{=} \text{elem } e \ x \ \parallel \ \text{elem } e \ xs \ \parallel \ \text{elem } e \ ys$$

$$\text{elem } e \ x \ \parallel \ \text{elem } e \ xs \ \parallel \ \text{elem } e \ ys$$

Lado derecho:

$$\text{elem } e \ (x:xs) \ \parallel \ \text{elem } e \ ys \stackrel{\text{def elem 1}}{=} \text{elem } e \ x \ \parallel \ \text{elem } e \ xs \ \parallel \ \text{elem } e \ ys$$

$$\text{elem } e \ x \ \parallel \ \text{elem } e \ xs \ \parallel \ \text{elem } e \ ys$$

Conclusión: Como el lado izquierdo y derecho dan lo mismo,

la T.I. queda probada

$$(P([]) \wedge \forall x :: a. \forall xs :: [a]. (P(xs) \Rightarrow P(x:xs))) \Rightarrow \forall ys :: [a]. P(ys)$$

Definiciones:

```
(++) :: [a] -> [a] -> [a]
{++} xs ++ ys = foldr (:) ys xs
```

```
preorder :: AT a -> [a]
{preorder} preorder = foldAT [] (\r i m d -> [r] ++ i ++ m ++ d)
postorder :: AT a -> [a]
{postorder} postorder = foldAT [] (\r i m d -> i ++ m ++ d ++ [r])
```

```
elem :: a -> [a] -> Bool
{elem0} elem _ [] = False
{elem1} elem e (x:xs) = e == x || elem e xs

foldr :: (a -> b -> b) -> b -> [a] -> b
{foldr0} foldr _ z [] = z
{foldr1} foldr f z (x : xs) = f x (foldr f z xs)
```

```
foldAT :: b -> (a -> b -> b -> b -> b) -> AT a -> b
{foldAT0} foldAT b f (Nil) = b
{foldAT1} foldAT b f (Tern v i m d) = f v (foldAT b f i) (foldAT b f m) (foldAT b f d)
```